



Republic of the Philippines  
**Laguna State Polytechnic University**  
Province of Laguna

**College of Computer Studies**

**ITEC 204 – Data Structures and Algorithms**

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- I. **Objective:** Write a short explanation of what your program is designed to do. (Minimum of 3 sentences)

**Answer:** This program, being an IT Automation Incident Ticket Manager made with the concepts of linear data structures, allows the user to organize and manage these so-called incident tickets being generated and are then manipulated through the user's inputs. Its functions include adding, displaying, searching, removing, and counting whatever data is in these tickets. All is done in a clean process, with room for error, and a bit of comedy on my part.

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- II. **Program Operations:** Briefly explain how your program performs each operation. (Insert Specific Code Line and Your Explanation)

**Add**

**Incident:**

```
def add():  
    inc_id = input("Incident ID: ").strip()  
    if find(inc_id):  
        print("ID already existing >:D."); return  
    bot = input("Bot: ").strip()  
    desc = input("Description: ").strip()  
    tickets.append({"incident_id": inc_id, "bot": bot, "description": desc})  
    print("Ticket has been added to what is basically a digital jail cell, wonderful.")
```

For the “Add” operation, it will first ask you to input the following data, the incident ID, bot, and description of the incident ticket you want to add. If ever the inputted ID is the same as one that is already in the system, it will notify that it already exists. Otherwise, it will be added through the format done in the append function, with the notification of success.

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```
def display():  
    if not tickets:  
        print("No active tickets. Must be on vacation."); return  
    for t in tickets:  
        print(f"{t['incident_id']:<15}{t['bot']:<20}{t['description']}")
```

For the “Display” operation, it simply checks if there are any active tickets inside of the system. If there are no active tickets, as seen in the if condition, then it will return a notification regarding it. If there are at least 1 or more active tickets, it will output all stored data from the system, displaying it all.

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#### Search

Incident:

```
def search():  
    t = find(input("Incident ID: ").strip())  
    print(t if t else "Weird, no ID found at all. Wait... look behind you...")
```

“Search” operation is simple enough. Finds and displays the specific incident ticket data after inputting the correct ID, sends back a notification if not correct ID. What is interesting to me is that you can integrate if-else conditions into the print function which is pretty cool.

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**Remove Resolved Incident:** (Switching text positions since font was changing, sorry po for the jarring transition) “Remove” operation, also very simple. User inputs the ID of the ticket that needs to be removed, the “find” looks for that ID, then removes the incident ticket data if it is the same value as the input. Otherwise, notification again (these explanations are getting redundant lol).

```
def remove():  
    t = find(input("Incident ID: ").strip())  
    if t:  
        tickets.remove(t); print("Ticket eradicated, disintegrated, annihilated.")  
    else:  
        print("Not found mate. If you wanna assassinate a ticket, be efficient. - Sniper from TF2")
```



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**Count Active Incidents:** Honestly, the “Count” operation is so simple that it looks boring. It literally just counts the total number of active tickets using the length function, that’s it.

```
def count():  
    print(f"Total active tickets: {len(tickets)}")
```

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- III. **Flowchart:** Create a flowchart showing the overall process and logic of your program.

Sadly, I cannot make the flowchart as of now, since I am currently taking care of my brother’s illness at his apartment. Without the materials I need at this point of time, I would need to leave out this part for now. Apologies po, Ma’am Max, and thank you po for understanding if ever.

- IV. **Conclusion:** Briefly explain what you learned from the activity and how the linear data structure was applied.

I learned a lot from this activity, especially since my prof during my first semester at LB did not really give us a good way to memorize each of these functions, and while I did learn from Sir Kaleb last semester, I did end up forgetting it all after the summer vacation. Anyways, I found new ways to use if-else conditions, methods and functions such as “find” and “strip”, and most importantly, the functions used in linear data structures like “append” and “remove”. I can see these being applied in systems like online menu websites, or even simple apps like calendars and calculators. Overall, a fun yet challenging activity that really needs a good amount of attention and time management to complete.

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